KAIRÓS

A KPI intelligence-to-action engine. It detects a material metric movement, decomposes it exactly, tests each explanation against a graded standard of proof, and recommends an action with a named owner — or refuses, and prices the cheapest experiment that would settle the question.

Kairós — the opportune moment. The distance between a number moving and someone acting on it is where the value sits.

Built for the **Accenture Innovation Challenge 2026 · BusinessIntelligence.ai**.

```
git clone https://github.com/aswanthiitm/kairos && cd kairos
./run.sh                      # installs, builds the estate, serves on :8000
```

No API key required. The engine runs complete without a language model and says so.

The one architectural commitment

> *Language models are strong at **proposing** causal explanations from world knowledge and near-random at **inferring** causation from correlation. So the model proposes and phrases. It never decides, and it never computes.*

Kıcıman et al. (TMLR 2024) found LLMs beat existing algorithms on knowledge-based causal tasks. The Corr2Cause benchmark (ICLR 2024) tested seventeen models on inferring causation from correlation and found them near random. Both are true. The architecture is a direct read of where the split falls.

It is enforced at runtime, not asserted:

- the model receives a **closed evidence packet** of already-computed facts, with no tools and no data access, so it cannot compute;
- afterwards a **numeric guard** parses every figure in its output and matches it against that packet. An unverifiable number fails the narrative, which is retried once and then replaced by the deterministic narrator.

Measured on a live run: **91–100% of reasoning steps are non-LLM**, and every step declares its own method and the reason that method was chosen.

Ten stages

STAGE	QUESTION	METHOD	REFUSES TO
SEMANTIC	Whose definition of this metric?	Rules · SQL	Compute a KPI two systems define differently without resolving the conflict
FITNESS	Is this estate fit to reason over?	Rules · SQL	Analyse an estate that fails the gate
DRIFT	Has the ground moved under the baseline?	Statistics · ML	Trust a learned ranker outside its training support
SIFT	Is the change real, and does it matter?	Statistics · Rules	Wake anyone for a statistically real but immaterial move
SPLIT	Where exactly?	Deterministic algebra	Leave a residual — the identity closes to under ₹1
SOURCE	Why, and how sure?	Causal · Statistics · Retrieval · ML	Award L3 when the placebo test fails
PROPAGATE	Where in the chain is the loss created?	Causal · SQL	Measure an upstream cause in its effect's window
FORECAST	What if nobody acts?	Statistics	Present an extrapolation in the same voice as a counterfactual
SOLVE	What should someone do?	Retrieval · Rules	Emit an action whose lever isn't in the contract
NARRATE	How do we say it to this reader?	LLM · Deterministic	Publish a number that isn't in the evidence packet

The Evidence Ladder

Bradford Hill's viewpoints (1965) operationalised on Pearl's causal hierarchy (CACM, 2019). **Nothing below L2 is ever phrased as a cause.**

RUNG	TEST	LANGUAGE PERMITTED
L0	co-movement	*"moved alongside"*
L1	precedence within the graph's declared lag, plus a dose–response gradient	*"associated with"*
L2	independent corroboration — sources not sharing a pipeline agree, and conflicting evidence is under 40%	*"likely cause"*
L3	counterfactual — difference-in-differences on the log-ratio series against a real untreated cohort, with a placebo test that must validate first	*"quantified at −20.6%, p<0.0001"*

The ladder is **not cumulative**. A validated counterfactual needs no text at all, so a persona denied CRM verbatims still reaches L3 on structured evidence.

Conflict is measured, not assumed. Evidence supporting a rival theme is counted; above a 0.4 ratio the hypothesis is marked *contested* and cannot reach L2.

CASE	ON-THEME	CONFLICTING	RATIO	VERDICT
Dispatch SLA	12	0	0.00	corroborated → **L3**
Price increase	2	10	**0.83**	**contested — rung suppressed**

The governed contract

Five YAML files, 545 lines. This is the surface a client tunes; the engine is portable across sectors without touching Python.

FILE	CARRIES
`kpi_contract.yaml`	Sources with grain, cadence and freshness SLA. Five KPIs with SQL, additivity, drivers, materiality thresholds, sliceable dimensions, lineage, access class. Six levers with owner, decision right, authority limit, reversibility.
`causal_graph.yaml`	Nodes with observables and type. Edges with sign, **lag in days**, and a mechanism an analyst can defend. Explicit blocks with reasons.
`entitlements.yaml`	Personas with row filters, denied columns, denied evidence domains, PII policy, narrative shape, delivery channel.
`playbooks.yaml`	Past interventions with **measured** outcomes — recovery rate, weeks to effect, cost, measurement method.
`llm_pricing.yaml`	Per-model prices, FX, and the routing policy.

Two rules the contract enforces absolutely: **an undeclared KPI cannot be computed**, and **an undeclared lever cannot be recommended.**

Semantic reconciliation

Finance and Operations define net revenue differently — one deducts returns, one also deducts shipping. Both are correct inside their own system. KAIRÓS resolves the conflict once, before analysis, records which definition won and under whose authority, and reports the gap. It does not silently pick one.

Fiscal calendar and hierarchy

Periods are computed on the declared **April–March Indian fiscal year**, not the Gregorian calendar. Dimensions carry real hierarchies — `region → city` — and the engine traverses them, so a regional movement drills to the cities that caused it and rolls back up exactly.

What is in the box

Five connected KPIs across four sources, three grains, four refresh cadences (60 / 1440 / 15 / 10080 minutes). They are causally connected, not merely co-located:

```
otd_pct ──10d──▶ reorder_rate_28d ──▶ order_volume ──▶ net_revenue
                                           ▲                ▲
                                           └──── asp ───────┘
                                   (identity: revenue = volume × price)
```

Five cases, each with ground truth planted in the generator so the engine can be *scored* rather than admired:

	CASE	PLANTED	VERDICT
S1	North revenue −18.8%	WH-3 dispatch SLA collapse → three named accounts cut reorder cadence at a 10-day lag, plus a tier-mix shift	**CONFIRMED · L3**
S2	West volume −8.9%	A competitor promotion and our own price rise, same day, same channel, no clean control	**COMPETING**
S3	New category, 19 days	Too short to fit seasonality	**INSUFFICIENT_HISTORY**
S4	WH-4 on-time delivery "improves"	Late-shipment rows silently stopped loading	**DATA_QUALITY**
S6	South modern trade −18.9%	Competitor promotion hitting one channel only — the other channels are an untreated control	**CONFIRMED · L3**

S2 and S6 are deliberate mirrors: the same class of cause, one confounded by design and one clean. The pair is what makes the abstention in S2 read as judgement.

Security

Entitlements are applied to the data and the evidence set **before anything reaches a prompt**. Post-filtering generated text is not a control an auditor will accept.

PERSONA	ROW	COLUMN	DOMAIN	RESULT ON S1
CFO	all	—	no CRM verbatims	L3 via counterfactual; 23 items withheld
RSM North	`region = North`	no margin	—	refused outright on a West query
Supply Chain	all	**no ₹ at all**	no verbatims	movement as % only; packet holds no currency figure
Analyst	all	—	—	full method detail and lineage

Three enforcement points: **pre-flight refusal** before any query runs, **row filters compiled into SQL**, and **packet-level redaction** so a denied figure cannot reach the model. A test asserts the Supply Chain packet contains no rupee value anywhere.

Decision rights are machine-checkable

Each recommendation is routed to a human–AI collaboration mode derived from evidence grade × value at risk against the owner's authority limit × lever reversibility.

RECOMMENDATION	MODE	WHY
Counter-promotion, ₹0.7 L	`AI_LED_HUMAN_APPROVES`	L3, reversible, inside authority
Service credit, ₹44.6 L	`HUMAN_LED_AI_SUPPORTS`	L3, but **exceeds the RSM's ₹25 L limit**
Price correction	`HUMAN_LED_AI_SUPPORTS`	hard to reverse
Under abstention	`HUMAN_ONLY_AI_ABSTAINS`	no cause established
—	`AI_DELEGATED`	**reserved and never assigned; a test enforces it**

Every lever is externally visible to a customer, moves money, or both. Full delegation is left empty on purpose, which makes EU AI Act Art. 14 human oversight an inspectable property of each recommendation rather than a policy sentence.

The learned layer

A gradient-boosted ranker (**written in numpy**, no ML framework dependency) trained on resolved historical episodes. Its authority is deliberately narrow, and stated in its own output:

> *advisory-reorder-only: the ML score may reorder candidates within the set the evidence ladder already admitted, and may not promote a rung, alter a verdict status, or add or remove a candidate.*

It exists because the hand-designed ranking multiplies by explanatory power, so a driver carrying no share of the movement — a competitor promotion, a price move — could never rank first however strong its evidence. The ranker learns that trade-off from outcomes. Holdout top-1 accuracy **0.60 against 0.32** for the heuristic.

The drift stage checks every scored candidate against the p01–p99 feature range recorded at training time. If too many fall outside, **the ranker's authority is withdrawn for that run** and the evidence heuristic ranks alone.

Estate operations

```
python cli.py --sweep
  scanned 73 slices → 24 material, 9 data-quality, 16 insufficient history, 24 suppressed
  suppressed because:
    persistence     20   did not persist for the required number of days
    statistical      8   inside the expected band once seasonality is removed
    pct_of_plan      4   movement is too small a share of the period plan
    abs_inr          1   below the rupee materiality floor in the contract
  grain-blocked: otd_pct × segment — measured at shipment grain, which carries no segment
```

Alert fatigue is a measured property, not a claim. The sweep also shrinks with entitlement: the analyst scans 73 slices, the North RSM 58.

```
python cli.py --backtest
  20 windows, 1 alert (rate 0.05), precision 1.00, recall 0.25
```

Twenty rolling windows across four quiet months, one alert, and it was the real event. Recall stays in the output: the event spans four windows and only became material in the last one. An engine that cannot report its own false-alarm rate is asking to be trusted on faith.

Cost, latency and scale

End-to-end, offline	**≈450 ms**
Repeat run (narrative cache hit)	**≈80 ms**, zero tokens, zero cost
Cost per insight	**₹0.30** — one call, ~2.5k in / ~200 out on Haiku 4.5
Non-LLM reasoning steps	**91–100%**

The system prompt is marked for provider-side prompt caching; the narrative cache keys on a hash of the evidence packet, so identical evidence for the same persona never pays twice.

Scale — `python scripts/benchmark.py --scales 1 10 100`, on a laptop:

ROWS	DAILY AGGREGATE	LATTICE SCAN	COHORT DID
45,065	0.3 ms	0.6 ms	0.4 ms
450,650	0.8 ms	0.9 ms	0.7 ms
4,506,500	**3.2 ms**	**2.2 ms**	**1.9 ms**

Those are the three queries SIFT, SPLIT and the counterfactual actually run. Everything else in a run is bounded by the segment and window, not by estate size.

The feedback loop

Analysts grade a diagnosis; the prior re-weights and the leader ranking changes on the next run. A **correction** — an analyst saying what the cause actually was — is stored as a labelled counter-example and surfaced on every later run of the same shape until someone amends the causal graph or a playbook. A recorded **outcome** re-estimates the playbook's effect size as a weighted mean, so expected-impact figures are calibrated by this company's own track record.

Learned state lives in `runtime/` and `cli.py --reset` clears it. A system that learns needs an explicit way to forget.

Running it

```
./run.sh                                      # web UI on :8000
python cli.py --scenario S1 --persona cfo     # headless single run
python cli.py --all                           # every case × persona
python cli.py --sweep --backtest              # estate operations
python cli.py --reset                         # clear learned state
python scripts/benchmark.py                   # scale curve
python -m pytest tests/ -q                    # 125 tests against planted ground truth
export ANTHROPIC_API_KEY=sk-...               # optional: live narration
```

Without a key the deterministic narrator runs. Two simulation modes exercise the full LLM path including token accounting; `simulate_bad` injects a fabricated figure so the numeric guard can be seen firing and falling back.

Interfaces

Web UI, CLI, and direct import of `pipeline.run()` share one engine. The UI is deep-linkable — every state is a URL — and carries a live feed ticker showing each source ageing and counting down to its next refresh.

API

`GET /api/meta` · `/api/freshness` · `/api/analyse` · `/api/sweep` · `/api/backtest` · `/api/semantics` · `/api/ml` · `/api/learning` — `POST /api/feedback` · `/api/outcome` · `/api/reset`

Platform mapping

Built custom on DuckDB so it runs anywhere with no account. The distinction the brief asks for:

CAPABILITY	CLASSIFICATION
Query execution	Custom-built on an embedded engine; maps to **native** warehouse SQL
Semantic contract & reconciliation	Custom-built; would be **configured** on dbt / Unity Catalog
Row / column / domain security	Custom-built; would be **native** on Snowflake or Unity Catalog
Text retrieval (BM25)	Custom-built; would be **native** via Cortex Search
Evidence ladder, mechanism ledger, delegation router	**Custom-built — no native equivalent exists**
Playbook memory	**Custom-built — no native equivalent exists**
Learned driver-ranker	Custom-built (numpy GBDT)
Narration	**Externally integrated** (Anthropic API), optional

The two rows marked *no native equivalent* are the product. We deliberately do not rebuild text-to-SQL or the semantic layer — Cortex Analyst, Databricks Genie and BigQuery Gemini were all GA by mid-2026. That layer is a platform feature now.

Repository

```
config/     semantic contract · causal graph · entitlements · playbooks · pricing
data/       seeded generator with planted ground truth; DuckDB estate
kairos/     31 engine modules, including ml/ — the learned ranker and its trainer
app/        FastAPI service and the single-page UI
tests/      125 tests scored against ground truth
scripts/    scale benchmark
docs/       business proposal · developer reference · demo video
design/     the interface design canvas
```

Limitations

Stated plainly, because the whole thesis is about not overclaiming.

- **Synthetic data.** Effect sizes are recovered because they were planted. On real data the honest expectation is more L1 and L2 and fewer L3s.
- **A counterfactual needs a real control.** Where every unit is treated, the engine correctly returns L1 and abstains. This is common.
- **The causal graph is hand-curated.** That is the point — it encodes domain judgement — but it is maintenance, and a wrong edge silently blocks a true cause.
- **BM25, not embeddings.** Fine at this scale; a production corpus needs a vector index.
- **Three seeded playbooks.** The asset compounds only with use.
- **The numeric guard is magnitude-tolerant** at 1% to survive rounding, so a fabricated figure landing within 1% of a real one can pass. It is a strong filter on material fabrication, not a proof of correctness.
- **Forecasting is trend continuation only** — no competitor response, no regime change. Beyond about four weeks it is indicative, not a number to plan against.
- **Delivery writes to an outbox and does not send.** There is no transport configured, and the code says so rather than implying otherwise.
- **Confidence is composed, not yet calibrated against realised outcomes.** The calibration machinery exists for the ranker; the recommendation confidence does not have enough closed loops behind it to claim

calibration.

Licence

MIT. Synthetic data only; no real customer data is included.